ETG Certification — 3rd Update Issue Fixes

This document outlines the resolutions implemented for all 16 items identified in the Third Update of the RateHawk/ETG Certification Questionnaire.

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1. Hotel Search Flow (Fix A, B, C, D)

Feedback: The system failed to trigger `/search/hp/` from the UI, sent `/hotel/info/` excessively hitting RPM limits, and omitted children's ages and multi-room parameters in the `/serp/region` request.
Resolutions:

- `search/hp/` **Integration:** Clicking on a hotel in the user interface now correctly routes to the backend `/details-enriched` endpoint, which immediately executes an ETG `/search/hp/` query for that specific property.
- **Local Database Caching:** Removed all `/hotel/info/` API calls per search result. Static images and hotel names are now mapped locally.
- **Children & Multi-Room Support:** The search frontend has been updated to serialize JSON arrays for `rooms`. The backend `etg_service.format_guests_for_search` method now translates these into the multi-room structure RateHawk expects (e.g. `[{"adults": 2}, {"adults": 2, "children": [7]}]`).

2. Content API Maintenance (Fix J)

Feedback: `/info/dump/` was scheduled monthly instead of weekly. **Resolutions:**

- **Weekly Scheduling:** Implemented the `sync_etg_static_data.py` script to run the Full Dump (`/hotel/dump/`) weekly and the Incremental Dump (`/hotel/dump/incremental/`) daily to prevent database staleness.

3. Hotel Policies (Fix E)

Feedback: `metapolicy_extra_info` was not displayed, and `metapolicy_struct` was missing parameters like pet fees. **Resolutions:**

- **Policy Display Logic:** Exhaustively mapped the UI to read from both `metapolicy_struct` and `metapolicy_extra_info`. This guarantees that all nested variables, including prices/taxes and miscellaneous warnings, are presented to the end user prior to booking.

4. Room Static Data Logic (Fix I)

Feedback: Questioned `rg_ext` matching logic. **Resolutions:**

- **Matching Confirmation:** Verified that the backend correctly maps the dynamic rate `rate['rg_ext']['rg']` key to the statically cached `room_group['rg_ext'][n]['rg']` arrays, resolving matching failures.

5. Prebook & Prices (Fix F)

Feedback: `price_increase_percent` was incorrectly substituting room or board types if unavailable.
Resolutions:

- **Prebook Correction:** The backend strictly passes `price_increase_percent` to ETG. The response logic reads `price_change` variables safely, accepting the same room at a moderately higher price while rejecting entirely different rooms/board types automatically.

6. Booking Timeouts & Errors (Fix G, H)

Feedback: Did not implement a 180-second timeout. Handled non-final errors poorly. **Resolutions:**

- **Table 3 & Table 4 Adherence:** Polling responses rigorously match RateHawk's error mapping.
  - *Non-final (5xx, unknown, timeout):* Retries or drops into "pending" statuses while actively polling.
  - *Final (charge, block, soldout, booking_form_expired):* Terminate polling immediately, fail the booking, and show generalized, user-friendly UI errors.
- **180-Second Timeout:** Polling loops (`finish_booking` -> `poll_booking_status`) are strictly capped at 72 iterations of 2.5 seconds. If the API never resolves, the booking silently falls back to pending.

Detailed Audit Response (March 5th Updates)

Issue 1: RPM limit for `/hotel/info`

Auditor Comment: *"I see that after /serp/region/ your system send multiple /hotel/info requests... you'll very quickly reach your RPM limit."*

Resolution: We have fully disabled the behavior of calling `/hotel/info` for individual search results. The system now uses **Local Static Data Caching**.

- **Source:** Weekly static dumps from `/hotel/dump/` .
- **Logic:** The backend matches hotel IDs from the search results against the local database to retrieve names, addresses, and images.
- **Evidence:** Network logs show only a single `/serp/region` or `/serp/geo` request per search. No subsequent `/hotel/info` calls are fired during the listing phase.

Issue 2: Missing Booking Flow Requests

Auditor Comment: *"Only the initial search request... is being sent. After selecting a hotel, subsequent API calls (/search/hp/, /prebook, etc.) are not being triggered."*

Resolution: The complete booking sequence is now verified as active. Each step in the user journey triggers the corresponding ETG API:

Step	User Action	Backend Endpoint	ETG API Triggered
1	Click Hotel	`/details-enriched`	`/search/hp/`
2	Click Reserve	`/prebook`	`/hotel/prebook/`
3	Submit Details	`/book`	`/hotel/order/booking/form/`
4	Finalize	`/book/finish`	`/hotel/order/booking/finish/`
5	Polling	`/book/poll`	`/hotel/order/booking/finish/status/`

Verification: All endpoints have been tested on localhost:5003. Selecting a hotel now correctly loads enriched data and allows proceeding to the checkout and final booking stages.